Real-Time PPE Compliance Monitoring on Construction Sites

A fine-tuned YOLO11 detector with temporal violation confirmation — concise technical report and system overview.

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Problem and Motivation

Hazard & scale

Construction sites are high-risk; missing hardhats and vests are frequent injury vectors. Manual checks are intermittent and labour-intensive.

Continuous monitoring opportunity

Fixed cameras can observe every worker continuously; computer vision can convert video into actionable safety signals.

Deployment gap

Most literature focuses on detection accuracy; fewer works cover decision rules, temporal filtering of transient errors, and operational review interfaces.

Objectives and Contributions

1. Detector: Fine-tune YOLO11m for 10 classes under a 0.70 h training budget on 2 GPUs.
2. Evaluation: Stratified validation/test splits; ablation with and without Test-Time Augmentation (TTA).
3. Error analysis: Per-class metrics and confusion pairs to prioritise safety-relevant errors.
4. System: Full stack — FastAPI backend, five-frame temporal confirmation, React dashboard, open-source release.

Related Work — Where we sit



Classical to modern

Prior studies span classical vision to modern YOLO-family detectors (Park et al., Fang et al., Nath et al., Wang et al.). Many report mAP but not operational decision logic.



Benchmark results

Benchmarks on CHV/CHVG datasets report strong mAP (≈86–90%) for top YOLO variants; however, they rarely describe alerting, temporal filtering or incident review UX.



Identified gap

We provide explicit rules turning bounding boxes into alerts, a temporal filter to suppress transient errors, and an integrated review interface for staff.

System Architecture — Pipeline & rules

Pipeline (high-level):

1. React dashboard (user controls)
2. FastAPI backend (session, incidents)
3. YOLO11m detector (per-frame detections)
4. Compliance layer (rule engine)
5. Temporal filter (5-frame confirmation)
6. Incident log → Dashboard

Detector settings

Confidence threshold default 0.40; NMS IoU 0.45 (adjustable 0.10–0.95 from UI).

Compliance rules

NO-Hardhat and NO-Safety Vest trigger incidents; NO-Mask advisory only. Compliant frame = ≥1 person + Hardhat + Safety Vest.

Temporal logic

Violation must persist for 5 consecutive frames to log an incident; 2 s cooldown prevents repeated alerts.

Dataset & Classes

Construction Site Safety Image Dataset (Roboflow, Kaggle) — pooled and re-split 80/10/10 stratified by dominant class (seed 42).

Split counts — Train: 2,240 images / 30,684 boxes; Validation: 280 / 3,554; Test: 281 / 4,114 — Total: 2,801 images, 38,352 boxes.

Class palette (10): Hardhat, Safety Vest, NO-Hardhat, NO-Safety Vest, Mask, NO-Mask, Person, Safety Cone, Machinery, Vehicle.

Note: heavy class imbalance — Person ≈ 9,872 boxes; vehicle ≈ 1,628; Mask ≈ 1,700.



Required PPE

Hardhat, Safety Vest



Violations

NO-Hardhat, NO-Safety Vest



Advisory

Mask, NO-Mask



Context

Person, Safety Cone, Machinery, Vehicle

Model & Training

Architecture

YOLO11m (Ultralytics) — 20.06M parameters, 68.3 GFLOPs @ 640 px. C3k2 backbone, SPPF, C2PSA attention, multi-scale neck, anchor-free head (strides 8/16/32).

Loss & optimisation

Loss weights: box reg 7.5, classification 0.5, distribution focal 1.5. Optimizer: SGD, LR 0.01, cosine schedule, batch 32, mixed precision.

Compute & budget

Kaggle — 2 × NVIDIA Tesla T4 (15 GB). 0.70 h budget → 75 epochs completed; best epoch 72. Augmentations: mosaic (disabled last 10 epochs), mixup, copy-paste, HSV, flip, scale.

Quantitative Results — Detection Performance

Key metrics (confidence default 0.40). Measurements are per-image; timings exclude I/O.

Split	mAP@0.5	mAP@0.5:0.95	Precision	Recall	ms/img
Validation	0.8679	0.6368	0.9236	0.8084	32.79
Test	**0.8659**	**0.6158**	**0.9135**	**0.8053**	**31.32**
Test (TTA)	0.8603	0.6206	0.9027	0.7853	85.98

Observations: validation/test parity suggests low overfitting; precision > recall indicates most errors are missed detections rather than false positives. TTA reduced mAP and is 2.7× slower — excluded from deployment.

Error Analysis & Temporal Impact

Per-class extremes

- Best AP: Machinery 0.9565, Mask 0.9432, Person 0.9250
- Weak: Safety Cone 0.7447, Hardhat 0.7975, Safety Vest 0.8273, NO-Mask 0.8055

Missed / confused cases

Small or thin objects: 16% Hardhat boxes missed; 14% of Safety Cone and Vest missed. Class swaps with safety impact: 9% NO-Hardhat → Hardhat (missed violation), 7% Vest → NO-Vest (false alarm), 10% Vest → Person (label drift).

Temporal filter effect

Five-frame confirmation suppresses transient single-frame errors but cannot remove persistent misclassifications. Temporal filtering reduces false incident volume and increases reviewer signal-to-noise.

Conclusion, Limitations & Future Work

Conclusion

YOLO11m achieves test mAP@0.5 = 0.8659, mAP@0.5:0.95 = 0.6158, inference ≈31.3 ms/img. Compliance rules + 5-frame confirmation produce reviewable incidents and reduce noise. ONNX export (80.5 MB) and open-source FastAPI + React stack provided.

Limitations

Single training run on one dataset split; no external site generalisation test. Compliance score is frame-level, not linked to individual workers. Temporal window design not tuned on annotated video.

Future work

1. Associate PPE boxes with person tracks for per-worker compliance and longitudinal reporting.
2. Evaluate across sites, weather, and lighting conditions; expand dataset diversity.
3. Tune temporal window on annotated video; improve small-object detectability (higher-res inputs, focal heads, augmentation focused on small objects).
4. Compare with transformer-based real-time detectors (e.g., RT-DETR) and multi-frame models.

Code & assets: github.com/parthmital/PPE-Compliance-Monitor